MANOBHI SRI RAM

Generative AI Engineer · Agentic AI Systems · LLM Orchestration · RAG Pipelines · LLMOps
Chennai, IN · +91 99199 38589 · LinkedIn · Portfolio · github.com/manobhisriram

PROFESSIONAL SUMMARY

Production-focused Generative AI Engineer with B.Tech in CS and 1 year of experience shipping end-to-end Agentic AI systems across healthcare, legal, and enterprise domains — adopted by 1,500+ employees across 12 countries, delivering 95% latency reduction, 94% cost savings, and 300+ hours automated monthly across 3 production systems. Deep expertise in multi-agent orchestration (LangGraph, AutoGen, CrewAI), RAG pipelines (Pinecone, Weaviate, Milvus, Qdrant), LLMOps (Langfuse, Datadog, Prometheus), fine-tuning (LoRA/QLoRA), and MCP server development. Deployed across AWS, Docker, and FastAPI with Claude, GPT-4o, Gemini, and open-source models.

Core Competencies: Generative AI, Agentic AI, LLM Orchestration, Retrieval Augmented Generation (RAG), LLM Fine-Tuning (LoRA/QLoRA), Prompt Engineering, Multi-Agent Systems, Vector Embeddings, AI Agents, Model Evaluation, LLMOps, MLOps, MCP Server Development, Python, REST API, FastAPI, AWS

WORK EXPERIENCE

ASCENDION | AI Engineer — Generative AI & ML · Chennai, TN *Aug 2025 – Present*

Project 1: Enterprise AI Media Intelligence Platform — Facial Recognition & Semantic Vector Search

Sole architect and developer of a company-wide facial recognition + semantic media retrieval platform, indexing 50K+ enterprise event photos across 1,500+ employees globally; presented live to company directors at every milestone.

- Achieved 97.3% facial match accuracy using a multi-model pipeline (DeepFace/FaceNet512, RetinaFace, MTCNN) with adaptive preprocessing for low-light, group, and varied-angle scenarios — deployed on AWS EC2 with zero downtime.
- Cut retrieval latency by 95% (8+ min → 0.4s) via Pinecone 512-dim cosine similarity vector DB with ThreadPoolExecutor concurrent batch inference.
- Reduced media processing time by 94% (13 hrs → 45 min); adopted by 1,500+ employees saving 200+ person-hours/month; privacy-first design with encrypted pipelines, zero persistent selfie storage, RBAC, and CloudWatch observability.

Project 2: Leadership Mobility Command Center — Generative AI + Agentic AI Enterprise Operations

Led a 4-person team as technical lead — designed and developed the entire GenAI and Python backend for an autonomous executive travel coordination platform across 12 countries.

- Built LangGraph multi-agent orchestration (flight intelligence agent, ground transport agent, shared trip-state memory) for fully autonomous executive travel coordination with zero manual intervention.
- Delivered a production RAG pipeline (LangChain · Claude 3.5 Sonnet · Qdrant) on 5,000+ vectorised trip records — eliminated all manual trip coordination saving 300+ coordinator hours/month; full LLMOps via Langfuse.
- Integrated Aviationstack telemetry, Google Maps, Microsoft Teams broadcasting, and Twilio/Firebase GPS for real-time end-to-end trip intelligence across 12 countries.
- Adopted as Ascendion's global executive mobility platform for all employees across 12 countries — presented and signed off by company leadership.

PROJECTS

Legal Contract Intelligence Platform — Agentic AI · Multi-Model · MCP · LLMOps *2026* GitHub

Reviews contracts in 47 sec for ₹0.78 vs 2–4 hrs of junior lawyer work (~₹1,400/review saved). 4-agent AutoGen pipeline · regulatory monitoring · Claude Desktop MCP.

- **4-agent AutoGen pipeline** across 5 models — Claude Sonnet 4.5 (extraction), GPT-4o (risk classification, 47-pattern library), Gemini 1.5 Pro (template comparison, 1M context), Claude Opus 4.5 (synthesis) at **$0.0094/review**; Claude Vision + GPT-4o fallback extracts tables, stamps, handwritten annotations, redlines across first 20 pages.
- **RAG on Milvus/Zilliz Cloud** (3 collections), OpenAI text-embedding-3-small, Cohere rerank-v3.0 (top-20→top-5, **+17.7% NDCG@5**), Groq Llama3, 34.2% cache hit rate; APScheduler SHA-256 hashes 5 Indian/EU regulatory sources daily, auto-maps compliance gaps to contracts.
- **4 MCP servers (17 tools)** for zero-API-knowledge Claude Desktop; LLM judge: **82.4% overall** (correctness 88.7%, calibration 80.5%, completeness 79.3%, actionability 81.2%) on 100 CUAD contracts; red-teaming drove failure rate **41%** → **9%** across 5 attack categories in 3 iterations.
- **LLMOps: Datadog** (ddtrace, LLMSpan APM, 40+ DogStatsD metrics, structlog) + Langfuse prompt replay; FastAPI (15 routes), PostgreSQL/Neon (8 tables, async SQLAlchemy + Alembic), Streamlit, Docker Compose, Render; 46 pytest functions with full LLM API mocking.

Enterprise AI Decision Intelligence Platform — Agentic AI · Governance · RAG · LLMOps *2026* GitHub

Enterprise AI governance — cost control, PII security, hallucination mitigation via 15-step pipeline. Saves ~₹1,40,000/year at 2,000 queries/day.

- **15-step query pipeline:** Presidio PII detection (Aadhaar/PAN/phone) → Groq classification (complexity/risk/debate, guaranteed JSON) → Weaviate semantic cache (0.92 threshold, **67.3% hit rate**) → guardrail matching → human-in-loop governance → Groq rewriting → Weaviate hybrid BM25+vector (top-10) → Cohere reranking (**+29.5% quality**).
- **LiteLLM complexity routing** (Haiku ₹0.08 / Sonnet ₹0.52 / Opus ₹2.31, auto-fallback) + long-term user memory injection; PostgreSQL (8 tables), async SQLAlchemy + Alembic, Unstructured (PDF/Word/Excel/PPT/CSV), permanent audit trail.
- **3-agent CrewAI debate:** Claude Sonnet (Advocate) · GPT-4o (Critic) · Gemini 1.5 Pro (Judge, 3 tools) — eliminates single-model bias on complex queries; **32.9% better decision quality** vs single-model, +71.4% improvement in opposing-perspective consideration.
- **Observability:** Langfuse per-stage tracing, Prometheus (8 metrics) + Grafana Cloud 8-panel dashboard; TruLens RAG Triad: **answer relevance 84.7%, context relevance 79.1%, groundedness 82.3%**; Promptfoo regression testing; FastAPI (18 endpoints), Docker Compose, Render + Streamlit Cloud.

MedAssist — Clinical Triage Agent (Healthcare AI) *2025* GitHub

Local-inference Agentic triage for Indian healthcare — symptoms in plain English → source-cited decision grounded in WHO/ICMR guidelines. Zero cloud dependency, full audit trail.

- **Fine-tuned Phi-3 Mini 3.8B** on 15,000 medical QA examples (ChatDoctor-HealthCareMagic-100k) via LoRA (r=4, alpha=8); training loss 1.921→1.219 in 1hr 57min on Kaggle T4; deployed via Ollama — patient data never leaves environment.
- **5-node LangGraph pipeline** (SymptomParser → SeverityAssessor → ClinicalRetriever → DrugChecker → TriageDecision); HIGH/EMERGENCY triggers parallel OpenFDA drug lookup; LOW/MEDIUM skips (~30% latency reduction); Pydantic v2 + instructor validation, auto-retry (max 3), graceful degradation.
- **Hybrid RAG on LanceDB** (vector + full-text, RRF fusion from scratch), semantic chunking (all-MiniLM-L6-v2, cosine 0.75) across 65 WHO/ICMR docs + live OpenFDA; context precision **0.65** → **0.85** vs fixed chunking.
- **Ragas eval:** faithfulness 0.92 · answer relevancy 0.90 · context precision 0.85 · context recall 0.87; 5/5 DeepEval assertion tests; GitHub Actions CI/CD blocks any push degrading faithfulness < 0.60 or causing severity misclassification; full Langfuse trace per decision; FastAPI + Streamlit · Hugging Face.

TECHNICAL SKILLS

Agentic AI and LLMs: LangChain, LangGraph, AutoGen, CrewAI, MCP Server Development, Claude (Anthropic), GPT-4o, Gemini, Groq, LiteLLM, Prompt Engineering, LLM Fine-Tuning (LoRA/QLoRA)

RAG and Vector Search: Retrieval Augmented Generation (RAG), Pinecone, Qdrant, Weaviate, Milvus, LanceDB, FAISS, Cohere Rerank, Hybrid Search (BM25 and Vector), Semantic Chunking, RRF Fusion, Vector Embeddings

LLMOps and Model Evaluation: Langfuse, Datadog, Prometheus, Grafana, Ragas, DeepEval, TruLens, Promptfoo, Weights and Biases, MLflow, Hugging Face, MLOps

Machine Learning and AI: Deep Learning, Natural Language Processing (NLP), Computer Vision, LLM Fine-Tuning, Phi-3, Ollama, OpenFDA Integration, AI Agents

Backend and Cloud: Python, FastAPI, REST API, PostgreSQL, Docker, AWS (EC2, S3, Bedrock, CloudWatch), Git, GitHub, Render, Neon, SQLAlchemy, Alembic

Languages and Frontend: Python, SQL, JavaScript, React, Streamlit

CERTIFICATIONS

- **Microsoft Certified: Azure AI Developer Associate** · Microsoft · 2026
- **AWS Certified Cloud Practitioner** · Amazon Web Services · 2023

EDUCATION

Vellore Institute of Technology (VIT), Chennai | B.Tech, Computer Science & Engineering *Aug 2021 – Aug 2025*

CGPA: 8.39 / 10

B.Tech Final Year Thesis — AI Based Post Harvest Management of Mushrooms · 2025

Best Project Award · First Author · Built entire ML/DL pipeline · 3-person team

- Built a custom dataset of 2,000+ mushroom images from scratch — no public dataset existed; images captured over multiple days with continuous freshness scores (1–90).
- Designed and trained the full ML/DL pipeline — custom CNN with augmentation achieved R^2 0.92, MSE 7.8, MAE 1.9 vs EfficientNet baseline of R^2 0.55; outperformed manual inspection (65%) and traditional image processing (75%).
- Deployed a real-time Streamlit interface for instant freshness prediction via image upload using TensorFlow, OpenCV, XGBoost, and Albumentations.